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Spatiotemporal Integrated Gradients: Contribution Method for Video Data

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ABSTRACT

Unfinished.

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1. INTRODUCTION

Suppose you have a convolutional neural network (CNN) model trained to predict class labels (digits, people, objects) and that it surpasses humans on such classification tasks. If this model is designed in the fashion of popular networks like ResNet-18, then these models would have hierarchical representations of visual inputs. Thus, simple features such as classical receptive fields would appear in early layers and complex, specific features like faces or a person would appear in later layers. This hierarchical representation of visual stimuli in models is well-founded and closely represents human level organization of feature representation in the ventral visual stream.(Necessary to cite the 30+ papers showing evidencing this)

The aforementioned logic assumes that the CNN has not learned a trivial look-up table for the inputs and their labels, which would make the following research questions and line of reasoning pointless. What representations within the CNN contribute to correct identification? What internal layer functions necessitate successful object categorization? And most importantly: what does each layer of the model see when given an input? Such line of questioning are at the core of explainable AI (XAI). XAI is closely related to the field of mechanistic interpretability, putting the 'science' in computer science and elucidating principles and laws of neural network computations. In simple terms, XAI and MechInterp have a shared goal of parsing how neural network models perform well-defined tasks.

1.1. *Background on Interpretability Methods*

For image classification tasks, interpretability tools like Gradient-weighted Class Activation Mapping (GradCAM) provides visual explanations for classification tasks. This method works by computing the gradients from an input with respect to the activations of the final convolutional layer, indicating the importance of each feature map. These gradients are averaged, summed, then thresholded to combine these features into a single map of important regions overlayed onto the original input. This method gives a visual representation of the input features that are most salient to the network. The XAI community has expanded on this method of interpretability in vision networks with GradCAM++, ScoreCAM, and Integrated Gradients. The last of which I will expand upon for the rest of this document.

Integrated gradients (IGs), is a gradient based method that calculates a path integral from baseline to stimulus to any layer of interest. This is accomplished through making a path, collecting gradients, approximating the integral from baseline to the stimulus. This explains which part of the network contributes to the output. Further, using integrated gradients enables network contribution mapping: identifying channel specific attributions for convolutional layers. By leveraging the fundamental theorem of calculus: the integral and differentiation are inverse operations; integrated gradients asserts linearity, sensitivity, and completeness. Thus, integrating gradients through a neural network is a more rigorous attribution method for explaining model computation and functions. The original method was shown to explain model choices in image, language, and molecular data–lending to it's broad applicability to explaining how networks arrive at their output.

1.2. *Motivation for Spatiotemporal Integrated Gradients*

We will expand the integrated gradient method in two ways: using (1) models trained to predict neuronal firing rates in response to (2) video inputs that the neurons observed. The following research questions can be asked: What does

each layer of the neural network observe as determined by contribution maps from top contributing channels? What cell types are necessary for emergent cell type phenomenology like direction selectivity, omitted stimulus response, or object motion detection?

The Integrated gradients method—while powerful, has been shown to work best on static datasets. This method accumulates gradients from baseline to the input, which allows the practitioner to glean what features of the input contribute most to model output. The question lies in what a good baseline should be for calculating this integral. In one case integrating from black, or zero, would be analogous to seeing what parameters of the model are salient as the full image comes into view. It can be argued that this approach biases the explanation, thus, averaging the path across many baselines (e.g. variable starting points between black and gray) up to the full image gives a better estimate of the path attribution.

Applying this method to videos would require a per-frame IG computation or averaging across batches of frames—which would result in computationally intensive and low quality estimations of the attributions. To sequentially compute integrated gradients accurately, we must compute a spatiotemporal integral. Sequences of frames must be processed in mini-batches, permitting both spatial and temporal investigation of network outputs to video data.

Given that spike trains and video inputs are time series data, it is important to know the trajectory of the contributions over sequences of events, allowing for better quality attributions in these data domains. Ideally, we want to maintain the original benefits of the standard integrated gradient application: completeness, sensitivity, and linearity. For example, if features from a sequence of frames have the same impact as the baseline input, then the resultant attribution value should be zero. However, adapting the original method to spatiotemporal data shows whether those features were truly ineffective at perturbing the model or if took a path that led to null result. The standard IG method does not allow us to observe the positive and negative cancellations that occur as a result of competing effects on model outputs. Additionally, accounting for the time allows us to elucidate time dependencies inherent to video and spike train data, preventing the independence assumption that state of the art methods use.

2. RELATED WORK

2.1. *Explainable Artificial Intelligence for Time Series Data*

In time-series data, the relationships between data points are influenced by their order and historical context. Naively applying the standard integrated gradient method and treating observations as independent, often fail to capture these dynamics, leading to the creation of Out-of-Distribution (OOD) intermediate inputs. Recent methods address this by incorporating temporal dependencies, primarily focusing on local feature masking or perturbation for final prediction attribution The following methods based on Integrated Gradients (IG) offer complementary approaches for using neural network models on time-dependent data. Recent efforts have applied IG to various sequential and multi-dimensional domains:

Temporality-Aware Integrated Gradients (1D): TIMING (Jang et al., 2025) explicitly addresses the OOD problem in $1D$ time series data, demonstrating that while conventional IG is theoretically sound, its naive application ignores temporal relationships. **TIMING** enhances IG by incorporating a temporal-aware path to improve directional attribution on data like medical records. In this extension they take subsamples of the time sequence data with which to compute the Riemann sum over a sufficient number of steps, then sum the attributions. While this method allows for higher quality temporal estimates of attribution, this method loses completeness as the section they sampled are not acquired from a linear path, but a randomly selected sequence of the history. This method addresses the OOD issue in the sacrifice of completeness.

Integrated Gradients for Spatio-Temporal Data (3D): Work on precipitation nowcasting (Ahangama et al., 2025) demonstrates the feasibility of applying standard **Integrated Gradients** to high-dimensional spatio-temporal image sequences (e.g., satellite video data) for region-of-interest identification in weather data to predict the next half hour of weather conditions. However, this method relies on the *simultaneous straight-line interpolation* path, which does not address the OOD problem inherent in video. Instead, they essentially apply the integrated gradient method over every Nth frame, accounting for time, but possibly ignoring the trajectory that the data took. This paper highlights that there may be an acceptable limit of estimation that does not require practitioners to compute gradients for the entire sequence of input frames. **Sequential Integrated Gradients (SIG) in NLP:** The work by Enguehard (2023) introduces **Sequential Integrated Gradients (SIG)** for language models, calculating token

importance by interpolating only the token of interest while *fixing the context* of all other words. This confirms the conceptual necessity of a sequential, context-preserving path for attribution in discrete, contextual data.

The existing use of the integrated gradient method reveals a critical distinction in objectives: most methods (masking, perturbation) focus on attributing features to the **final prediction score**. Our approach, the **Piecewise Spatiotemporal Integrated Gradients (SIG)**, is novel because we combine the necessity of a causally-constrained path with attribution to **intermediate channel units** in a biologically-constrained model. This addresses two gaps simultaneously:

Methodological Gap: We overcome the OOD and causal inconsistency of naive IG on video by defining a **piecewise linear path** that fixes the history frames $\mathbf{x}_1$ to $\mathbf{x}_{t-1}$ when integrating frame $\mathbf{x}_t$. This ensures the attribution of each time step is measured within a temporally coherent context.

Scientific Gap: The core objective is not merely prediction attribution, but to leverage the model as a platform for neuroscientific inquiry. By performing **Output-to-Hidden** attribution, we investigate whether the learned internal units of the model trained to predict V1 spike trains have **temporal correspondence** to known biological phenomenology.

This framework is essential for probing functions that are inherently time-dependent in the visual system. This functional identification is critical because many foundational operations in the visual system are inherently **time-dependent** and thought to arise from the interaction of distinct cell types. For example, phenomena like **polarity reversal**, **latency encoding**, **frequency doubling**, **motion anticipation**, **object motion detection**, and **omitted stimulus response** emerge from the precise spatiotemporal summation and interaction of inputs from various upstream pathways. Our approach provides a tool to extract the temporal contribution map from many cell types simultaneously, which is essential to explain the emergence of these complex, predictive, and time-dependent phenomena.

3. METHODS

3.1. *Standard Integrated Gradients*

The standard integrated gradients method computes attributions for static images by integrating gradients along a path from a baseline to the input. The attribution for feature i is given by:

$$\mathrm{IG}_i(x) = (x_i - x_i') \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} d\alpha \tag{1}$$

where F is the model, x is the input image, x' is the baseline (typically a black image), and α parameterizes the path from baseline to input

Algorithm 1 Standard Integrated Gradients (IG) Method

Require: F: model; x: input image; c: target neuron/class; x': baseline (e.g., black image); n: number of interpolation steps
Ensure: A: attribution map showing spatial importance

Initialize $A \leftarrow \mathbf{0}$ with same shape as x
Generate interpolation coefficients $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ evenly spaced in $[0, 1]$
for each α in $\{\alpha_1, \ldots, \alpha_n\}$ **do**
    Compute interpolated input: $x_\alpha \leftarrow x' + \alpha \times (x - x')$
    Enable gradient tracking for x_α
    Forward pass: $y \leftarrow F(x_\alpha)$
    Select target output: $y_c \leftarrow y[c]$
    Backward pass to obtain gradients: $g_\alpha \leftarrow \frac{\partial y_c}{\partial x_\alpha}$
    Store g_α
end for
Approximate path integral using trapezoidal rule:
    $IG \leftarrow 0$
for $i = 1$ to $n - 1$ **do**
    $IG \leftarrow IG + 0.5 \times (g_i + g_{i+1})$
end for
Compute mean integrated gradient: $IG_{\text{mean}} \leftarrow IG/n$
Compute final attribution: $A \leftarrow (x - x') \odot IG_{\text{mean}}$
return A

3.2. *Spatiotemporal Integrated Gradients*

For video sequences, we introduce Spatiotemporal Integrated Gradients (SIG), which extends the standard IG method to handle temporal data. The key insight is to compute a piecewise path where each frame is revealed sequentially. For a video with T frames, the attribution for feature i in frame t is:

$$\mathrm{SIG}_{i(t)}(x) = \sum_{k=1}^{T} \mathrm{IG}_{i(t),k}(x) \tag{2}$$

where each segment contribution is given by:

$$\mathrm{SIG}_{i(t),t} = (x_{i(t)} - x_{i(t-1)}) \times \int_{(t-1)/T}^{t/T} \frac{\partial F(r^*(\alpha))}{\partial x_{i(t)}} \cdot T \, d\alpha \tag{3}$$

3.2.1. *Defining the Piecewise Path*

We define the path $r^*(\alpha)$ for $\alpha \in [0, 1]$, where the total path is divided into T equal-length segments, each of length $1/T$.

Let $\mathbf{X}_t^{\text{partial}}$ be a "partial input" where the first t frames are set to their true values $(x_1, \ldots, x_t)$ and the remaining $T - t$ frames are set to their baseline values $(x'_{t+1}, \ldots, x'_T)$. We define $\mathbf{X}_0^{\text{partial}} = \mathbf{X}'$.

For the t-th segment, corresponding to $\alpha \in \left[\frac{t-1}{T}, \frac{t}{T}\right]$, we transition from the partially revealed input $\mathbf{X}_{t-1}^{\text{partial}}$ to the next partially revealed input $\mathbf{X}_t^{\text{partial}}$.

The scaled parameter β within the segment $[0, 1]$ is defined as:

$$\beta = \alpha \cdot T - (t - 1), \quad \text{for } \alpha \in \left[\frac{t-1}{T}, \frac{t}{T}\right] \tag{4}$$

The path $r^*(\alpha)$ is then defined by:

$$r^*(\alpha) = \mathbf{X}_{t-1}^{\text{partial}} + \beta \cdot (\mathbf{X}_t^{\text{partial}} - \mathbf{X}_{t-1}^{\text{partial}}) \tag{5}$$

Algorithm 2 Spatiotemporal Integrated Gradients (SIG) Method

Require: F: model; X: input video with T frames; c: target neuron/class; X': baseline video (all black frames); s: number of interpolation steps per segment

Ensure: Spatiotemporal attribution tensor $\mathrm{SIG}_{i(t)}$ for each feature i in frame t

Initialize $X' \leftarrow$ video of T black (zero) frames
Define partial input states: $X_0^{\text{partial}} \leftarrow X'$, $X_T^{\text{partial}} \leftarrow X$
Initialize storage for frame-wise attributions $\mathrm{SIG}_{i(t)}$
for $t = 1$ to T **do**
  Define segment boundaries on the integration path: $\alpha \in [\frac{t-1}{T}, \frac{t}{T}]$
  Frames $1, \ldots, t-1$ fixed at true values, frame t transitions from baseline to true value, frames $t+1, \ldots, T$ remain baseline.
  Generate s interpolation steps $\{\beta_1, \beta_2, \ldots, \beta_s\}$ where $\beta_j = \frac{j}{s}$
  for $j = 1$ to s **do**
    Construct partial video $X^{\text{partial}}(j)$:
    For frames $1, \ldots, t-1$: $x_k^{(j)} = x_k$
    For frame t: $x_t^{(j)} = x'_t + \beta_j \times (x_t - x'_t)$
    For frames $t+1, \ldots, T$: $x_k^{(j)} = x'_k$
    Stack all frames as required by F (e.g., concatenate along channel dimension)
    Forward pass: $y \leftarrow F(X^{\text{partial}}(j))$
    Select target output: $y_c \leftarrow y[c]$
    Backward pass to obtain gradients: $g_t^{(j)} \leftarrow \frac{\partial y_c}{\partial x_t^{(j)}}$
    Store $g_t^{(j)}$
  end for
  Approximate integral for frame t using trapezoidal rule:
  $\mathrm{IG}_t \leftarrow 0$
  for $j = 1$ to $s-1$ **do**
    $\mathrm{IG}_t \leftarrow \mathrm{IG}_t + 0.5 \times (g_t^{(j)} + g_t^{(j+1)})$
  end for
  Scale by segment length: $\mathrm{IG}_t \leftarrow T \times \frac{\mathrm{IG}_t}{s}$
  Compute frame attribution:
  $\mathrm{SIG}_{i(t)} \leftarrow (x_{i(t)} - x'_{i(t)}) \odot \mathrm{IG}_t$
  Store $\mathrm{SIG}_{i(t)}$ in attribution tensor
end for
return Complete attribution tensor SIG containing spatiotemporal importance scores satisfying $F(X) - F(X') = \sum_{t,i} \mathrm{SIG}_{i(t)}$

3.3. *Contribution Map Visualization*

After computing attributions using either IG or SIG, we can identify which channels in a given layer contribute most to the model's predictions. This enables visualization of what the model "sees" at different layers.

4. RESULTS

4.1. *STIG Reduces to Standard IG: A Mathematical Proof*

The STIG method initially appears to address the temporal dependencies inherent in video data through its piecewise sequential path construction. However, a rigorous mathematical analysis reveals that this approach does not escape the fundamental constraint imposed by the path independence theorem.

4.1.1. *Theorem: Path Independence of Integrated Gradients (Lifted from Harvard-21A)*

For any continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ and any piecewise smooth path $\gamma : [0, 1] \to \mathbb{R}^n$ connecting baseline x' to input x:

$$\int_0^1 \nabla f(\gamma(\alpha)) \cdot \gamma'(\alpha)\, d\alpha = f(x) - f(x') \tag{6}$$

This result is independent of the specific path taken.

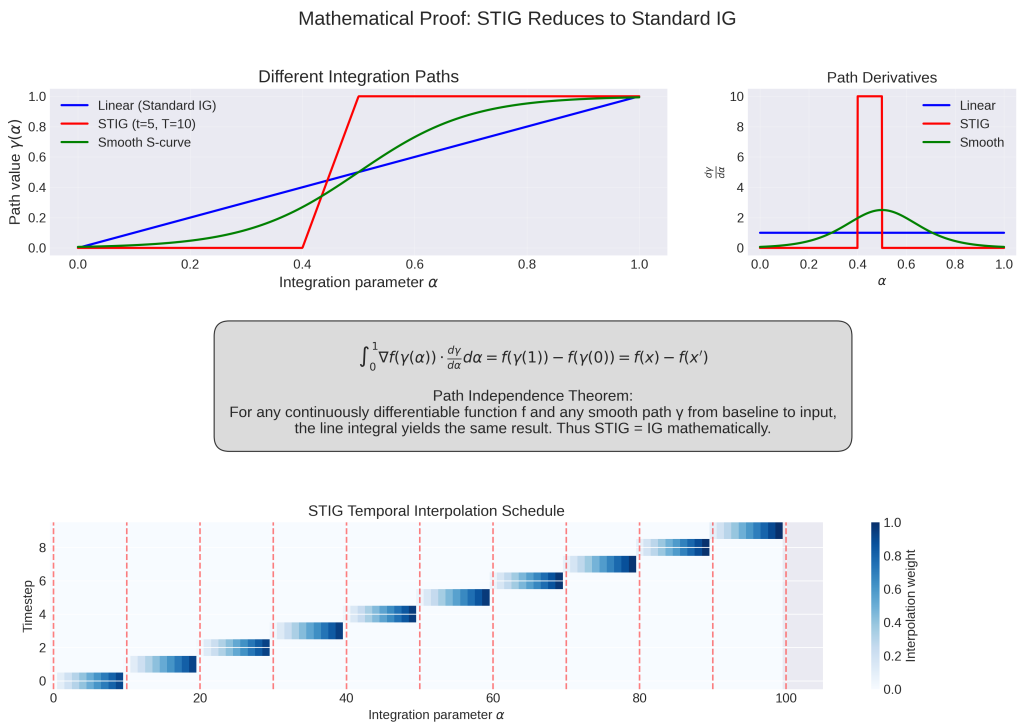

Figure 1. Path Independence of Integrals

A vector field $\vec{F} = \langle P, Q \rangle$ is called **path independent** if for any two points A, B and any two paths C_1, C_2 connecting A with B, the line integrals agree: $\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$. A vector field $\vec{F}$ has the closed loop property or is called **conservative** if for any closed path C the line integral is zero: $\oint_C \vec{F} \cdot d\vec{r} = 0$.

Conservative implies path independence. Two paths C_1, C_2 from A to B define a closed loop C which can be written as $C_1 - C_2$, meaning C_2 is traversed backwards. The integral along C is zero implies that the line integrals along C_1 and C_2 agree.

Path independence means gradient field. If f is a function, then $\vec{F} = \nabla f = \langle f_x, f_y \rangle$ is called the **gradient** of f. We can construct a function $f(x, y) = \int_{C(x,y)} \vec{F} \cdot d\vec{r}$, where $C(x, y)$ is a path from $(0, 0)$ to (x, y). The function is independent of the path. To check $f_x = P$, choose a path going from $(0, 0)$ to $(0, y)$ to (x, y), then check $f_x(x, y) = P(x, y)$ at this point. To verify $f_y = Q$, choose a path going from $(0, 0)$ to $(x, 0)$ and from there to (x, y).

Gradient fields are irrotational. If $\langle P, Q \rangle = \langle f_x, f_y \rangle$, then $Q_x - P_y = f_{yx} - f_{xy} = 0$ by Clairaut's theorem.

Irrotational implicates conservative. Given a closed loop C bounding a region G, Green's theorem implies $\oint_C \vec{F} \cdot d\vec{r} =_G \operatorname{curl}(\vec{F})\, dA = 0$.

Therefore, if $\vec{F}$ is defined everywhere, the following four properties are equivalent: 1) $\vec{F}$ is path independent, 2) $\vec{F}$ is conservative, 3) $\vec{F}$ is a gradient field, 4) $\vec{F}$ is irrotational (1).

4.1.2. *Application to STIG*

The STIG method attempts to use a temporally-aware piecewise linear path that interpolates timesteps sequentially. However, when we compute the integral:

$$\text{STIG}_{i,t} = (x_{i,t} - x'_{i,t}) \times \int_0^1 \frac{\partial f(\gamma(\alpha))}{\partial x_{i,t}} \cdot \frac{\partial \gamma_{i,t}(\alpha)}{\partial \alpha}\, d\alpha \tag{7}$$

Through change of variables and the fundamental theorem of calculus, this reduces to:

$$\text{STIG}_{i,t} = (x_{i,t} - x'_{i,t}) \times \int_0^1 \frac{\partial f(x' + \beta(x - x'))}{\partial x_{i,t}}\, d\beta = \text{IG}_{i,t} \tag{8}$$

Even though the path derivatives appear qualitatively different across interpolation schemes (Figure 1), with some paths moving rapidly through α space (exhibiting large derivatives over short intervals) while others progress more gradually (maintaining smaller derivatives over extended intervals), their contributions to the line integral necessarily balance. This occurs because the total path displacement from baseline to input remains invariant across all continuous paths. Consequently, despite the intuitive appeal of a temporally-structured interpolation schedule, the STIG formulation does not resolve the out-of-distribution artifact endemic to time-series data when applying gradient-based attribution methods.

4.2. *The Architecture Problem: Static Spatiotemporal Receptive Fields*

4.2.1. *Motivation: Dynamic Vision in Biological and Embodied Systems*

Understanding how visual systems process dynamic, naturalistic stimuli is fundamental to both computational neuroscience and embodied robotics. In our research linking biologically-constrained neural networks trained on V1 recordings to quadruped robot control, we require models that capture the temporal dynamics of visual processing. Biological visual cortex exhibits rich temporal dynamics in response to naturalistic stimuli—neurons show adaptation, prediction, motion anticipation, and context-dependent responses that unfold over time. Similarly, embodied robotic systems operating in real-world environments must process continuous visual streams where temporal relationships drive behavioral decisions.

To validate that our biologically-constrained models capture these dynamics, and to understand how they generate control signals for robotic locomotion, we need interpretability tools capable of explaining how models process and are changed by dynamic inputs. This requires moving beyond characterizing static receptive fields to understanding the temporal evolution of visual processing. However, as we demonstrate below, the combination of our current architecture and attribution method fundamentally limits our ability to probe these dynamics.

4.2.2. *The Core Issue with Conv2D on Temporally-Stacked Inputs*

When using Conv2D architectures with the temporal dimension encoded as input channels, the model learns a fixed spatiotemporal receptive field (STRF). This architectural constraint creates fundamental problems for understanding temporal processing:

First, the model exhibits *fixed temporal sensitivity*, maintaining a rigid temporal weighting structure that cannot adapt to input content. In biological systems, temporal sensitivity is highly dynamic—neurons adjust their temporal integration windows based on stimulus statistics, behavioral context, and attention. Second, there exists *no causal structure* in the computation: all timesteps are processed simultaneously without any notion of temporal order or precedence. This violates the fundamental causality of both biological vision and embodied systems, where future information cannot influence past processing. Third, and most critically for interpretability, this creates *attribution ambiguity* where we cannot distinguish between important temporal events in the input, instances where the input happens to align with the filter's fixed sensitivity pattern, and the model's inherent architectural biases.

For naturalistic stimuli—the diverse, time-varying visual inputs an animal or robot encounters—this ambiguity becomes particularly problematic. We cannot determine whether high attribution to a particular temporal window reflects a meaningful feature in that moment (analogous to a biological neuron responding to motion onset) or merely the alignment of arbitrary input with the model's fixed temporal filter. This fundamentally undermines the use of integrated gradients as an explainability method for understanding how biologically-constrained models process dynamic, naturalistic visual streams.

4.2.3. *Mathematical Formulation of the Problem*

For a Conv2D model treating time as channels, the output can be expressed as:

$$y = \sum_{t=1}^{T} W_t * X_t + b \tag{9}$$

where W_t is the learned weight for temporal position t. The gradient-based attribution becomes:

$$\frac{\partial y}{\partial X_t} = W_t \tag{10}$$

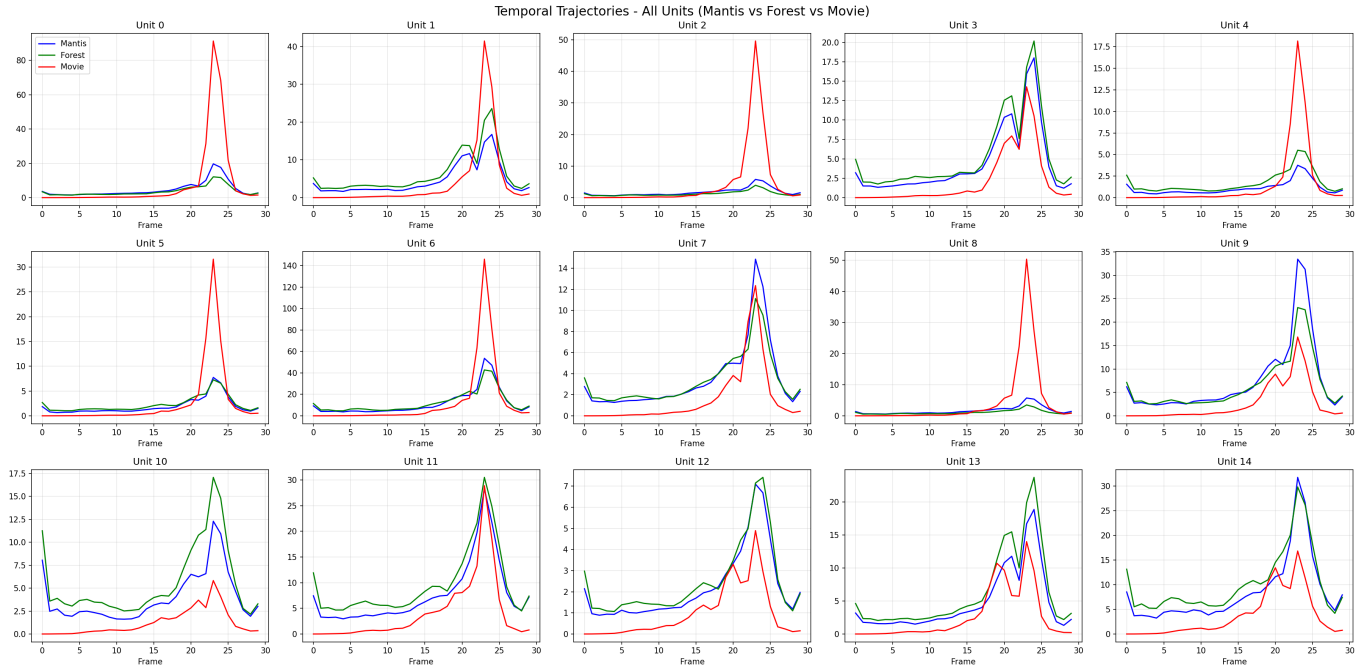

Figure 2. Standard integrated gradient result demonstrating the temporal filter bank learned from the mouse retina. Two repeated, static images (Forest-Green and Mantis-Blue) are compared to a movie input of a drifting bar-Red

This gradient is input-independent for the temporal structure. The attribution will always show the same temporal pattern scaled by the input magnitude:

$$\text{Attribution}(t) \propto X(t) \cdot W(t) \tag{11}$$

Given the batch processing implementation of integrated gradients, we are essentially chunking our sequence and asking how the fixed gradient output corresponds to each chunk. The fixed nature of the STRF results in attributions that average out between different inputs and become highly correlated with one another, regardless of the dynamic content being processed. The mathematical formulation above reveals that the Conv2D-on-time architecture functions as a filter bank of spatiotemporal receptive fields.

4.2.4. *Implications for Biological and Embodied Systems*

This analysis reveals a crucial distinction for interpreting biologically-constrained models. Integrated gradients reveals where the model looks through its learned receptive field structure, but cannot explain how specific dynamic inputs drive specific responses through that structure. For our Conv2D-on-time architecture trained on V1 data, integrated gradients reveals the fixed STRF template the model has learned, but cannot explain why one naturalistic video sequence produces one pattern of predicted neural activity while another sequence produces a different pattern.

In the context of biological vision, this limitation manifests as the ability to characterize a model unit's receptive field—useful for comparing learned representations to known V1 cell types—without understanding how different temporal features in naturalistic stimuli drive different response dynamics through that receptive field. For robotics applications, we can identify what spatiotemporal patterns the vision encoder is sensitive to, but cannot explain how different dynamic scenes lead to different control outputs. This is analogous to knowing the prescription of glasses (the architectural bias) but being unable to explain what the wearer currently sees through those glasses as the visual world unfolds.

This fundamental constraint is revealed most clearly through computing integrated gradients on repeated, static frames, where the method's assumptions align with the data structure. To understand causal, dynamic relationships in naturalistic visual processing—whether for modeling biological vision or controlling embodied robots—we require either different attribution methods (such as temporal attention mechanisms that explicitly model dynamic feature

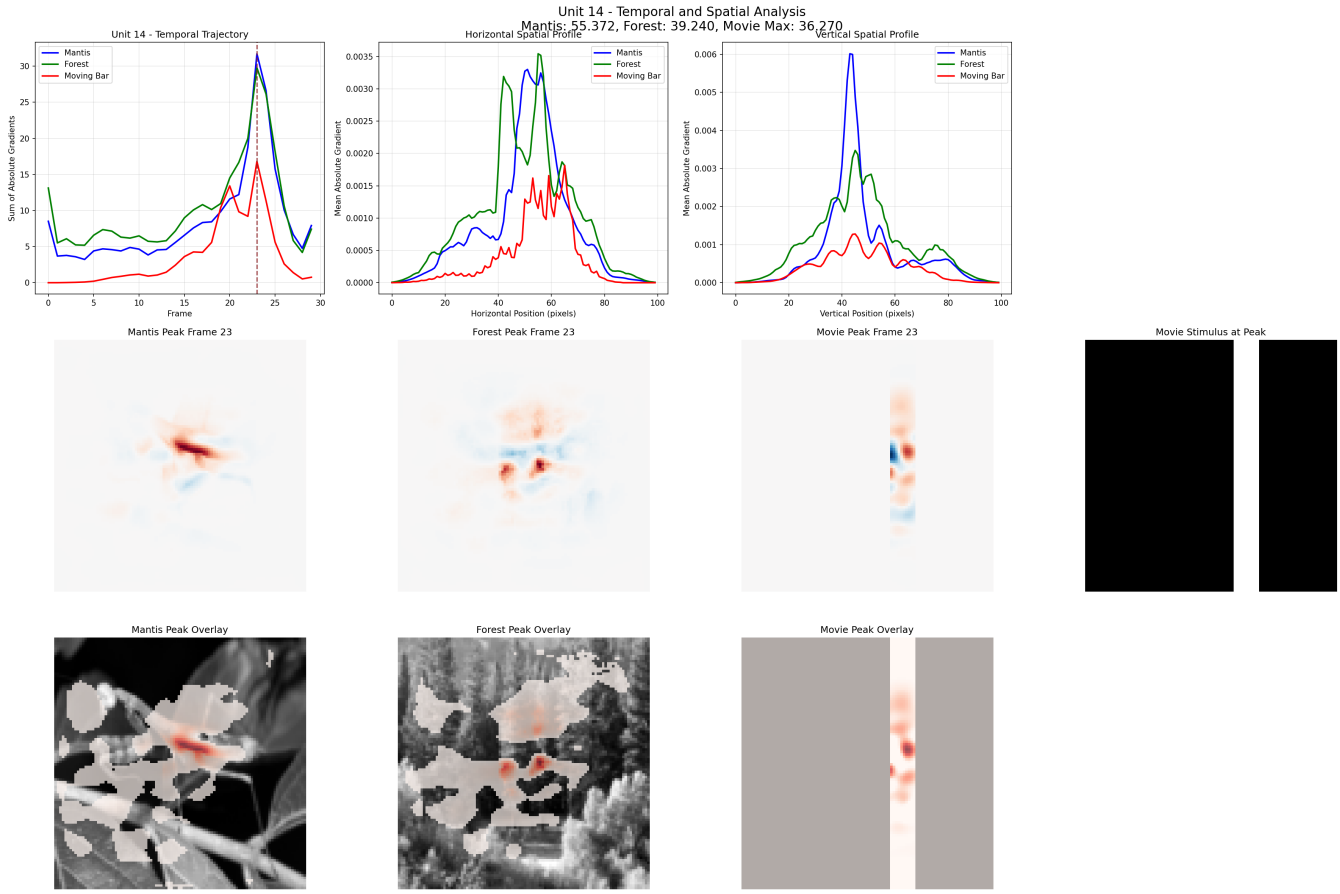

Figure 3. Example of spatiotemporal filter or static and motion images

routing), different architectures (that avoid fixed temporal filters in favor of recurrent or state-dependent processing), or both approaches in combination.

The integrated gradients method can be more accurately framed as a bias detector rather than a tool for dynamic explainability. This limitation finds support in the method's history of successful application to images, molecules, and single sentences—all domains characterized by fixed, static inputs. The method's theoretical foundation and empirical success naturally align with applying a fixed analytical tool to fixed input structures, but this alignment breaks down when extended to the inherently dynamic, temporal domains that characterize both biological vision and embodied robotic control.

5. DISCUSSION

5.1. *Temporal Out-of-Distribution Artifacts and Attribution Distortion*

5.1.1. *The OOD Problem in Time Series*

Temporal data presents unique out-of-distribution challenges that fundamentally differ from static data domains (2). Distribution shifts occur over time in financial markets, neural responses, and natural systems including video sequences, creating non-stationary behavior that violates standard machine learning assumptions. Temporal dependencies arise because future predictions depend on past context in ways that break the independent and identically distributed (i.i.d.) assumption underlying many attribution methods. Most critically, causal structure matters fundamentally in temporal data: the order of events carries information that cannot be captured by treating timesteps as interchangeable features.

5.1.2. *How Static STRFs Cause Attribution Distortion*

The fixed STRF architecture creates systematic attribution distortions that obscure true causal relationships. Consider two contrasting scenarios: in Scenario A, a spike in neural activity at $t = 5$ causes a downstream response at $t = 15$ through temporal integration and circuit dynamics. In Scenario B, a spike occurs at $t = 15$ with no causal relationship to earlier events. A Conv2D model with time-as-channels cannot distinguish these fundamentally different cases. Both scenarios will produce similar attributions if the learned STRF has high sensitivity at $t = 15$, regardless of whether that sensitivity reflects true causal mechanisms or merely correlational patterns in the training data.

This attribution distortion becomes particularly problematic when interpreting model units as analogs of biological neurons. In real neural circuits, the timing of inputs relative to one another determines synaptic integration, dendritic computation, and ultimately spike generation. A model that treats all temporal positions simultaneously cannot capture these dynamics, and attribution methods applied to such models cannot recover the underlying causal structure. The attributions reveal only where the fixed filter is sensitive, not how temporal patterns in the input drive temporal patterns in the output.

5.1.3. *The Solution: Architectures with Temporal Causality*

Models designed for temporal data must satisfy several architectural constraints to enable meaningful attribution. First, they must respect temporal order by processing information sequentially or with explicit causal constraints that prevent future information from influencing past computations. Second, they must exhibit adaptive temporal sensitivity, learning context-dependent temporal importance rather than applying fixed temporal weights. Third, they must maintain state by carrying information forward through time, allowing the model to build representations that integrate over temporal context.

Alternative Architectures—Three-dimensional convolutions treat time as a proper dimension with learnable temporal kernels, enabling the model to learn hierarchical temporal features while maintaining some notion of temporal locality. Unlike Conv2D with time-as-channels, 3D convolutions can learn different temporal filters at different spatial locations and depths in the network, providing greater flexibility in capturing spatiotemporal patterns.

Recurrent networks such as Long Short-Term Memory (LSTM) or Gated Recurrent Units (GRU) explicitly model temporal dependencies through hidden states that evolve over time. These architectures possess natural causal structure—the hidden state at time t depends only on current input and past hidden states, ensuring the model cannot access future information. The gating mechanisms in these networks will provide adaptive temporal sensitivity, allowing the model to learn which temporal contexts are relevant for different inputs.

Temporal Convolutional Networks (TCNs) employ causal convolutions to ensure proper temporal order while using dilated convolutions to capture long-range dependencies efficiently. This architecture maintains computational advantages similar to standard CNNs while respecting the fundamental causality constraints of temporal data. The exponentially growing receptive fields created by dilation allow the network to integrate information across long temporal spans without the vanishing gradient problems that can affect recurrent architectures.

5.1.4. *Implications for Attribution Methods*

When Standard IG is Sufficient—When the model architecture properly encodes temporal causality through recurrent connections, causal convolutions, or explicit sequential processing, gradient-based attribution methods can correctly capture temporal importance. In these cases, temporally-aware modifications to integrated gradients such as TimeX++ (7) and TIMING (8) can provide meaningful explanations of how temporal patterns in the input drive model predictions. The attributions reflect true model behavior because the architecture itself respects the causal structure of the temporal domain.

Fundamental Limitations of Attribution—No attribution method can recover meaningful temporal importance if the underlying architecture does not support temporal reasoning. When models use static temporal weights as in Conv2D with time-as-channels, when temporal causality is not built into the architecture, or when the model treats all timesteps simultaneously rather than sequentially, attribution methods can only reveal the model's fixed biases rather than dynamic input-output relationships. The problem lies not in the attribution method but in the architectural constraints that prevent the model from learning true temporal dynamics.

For modeling retinal ganglion cell responses to video stimuli and V1 neural dynamics during naturalistic vision, we must replace Conv2D architectures that treat time as input channels with architectures that respect temporal order.

Conv3D convolutions or hybrid Conv-LSTM architectures can capture true spatiotemporal dynamics by allowing temporal receptive fields to evolve with network depth and by maintaining recurrent state across time. Once the architecture properly encodes temporal causality, temporally-aware integrated gradients methods become sufficient for explaining model behavior without requiring fundamentally new attribution frameworks.

For embodied robotic systems operating in dynamic environments, these architectural considerations become equally critical. A robot's vision system must process temporal information causally—decisions at the current timestep can only depend on current and past observations, never future ones. Sequential architectures that maintain temporal state allow the system to build predictive representations that anticipate future events based on temporal context, mirroring the predictive processing observed in biological visual systems. The ability to interpret these temporal dynamics through attribution becomes essential for validating that biologically-constrained models capture the relevant computational principles for deployment in robotic control.

6. CONCLUSION

6.1. *Defining the Research Question*

Our analysis addresses a fundamental question in explainable AI for temporal data: *Can attribution methods like integrated gradients meaningfully explain how neural network models process temporal dynamics in video data, and what architectural properties are necessary for such explanations to reflect true causal temporal processing?*

This question decomposes into three specific components:

What is causal structure in temporal sequences? In a temporal sequence of video frames, causal structure refers to the constraint that information flows unidirectionally through time—an event at time t can influence events at time $t + \Delta t$ (where $\Delta t > 0$), but not vice versa. For neural responses to visual stimuli, this manifests as temporal integration windows, where a neuron's response at time t depends on stimuli presented at times $t - \tau$ through t (for some integration window τ), but cannot depend on future stimuli at $t+1$ or later. In naturalistic vision, this causality is fundamental: motion direction, temporal prediction, and adaptation all require that the visual system process information in temporal order.

How should causal structure be encoded in network architectures? An architecture "accounts for" causal structure when its computational graph enforces temporal ordering constraints. Recurrent architectures (LSTMs, GRUs) achieve this through hidden states that evolve sequentially: $h_t = f(x_t, h_{t-1})$. Causal convolutional architectures achieve this by restricting receptive fields such that the output at time t depends only on inputs at times $\leq t$. In contrast, Conv2D with time-as-channels violates causality by computing $y = \sum_{t=1}^{T} W_t * x_t$, where all temporal positions contribute simultaneously with fixed weights W_t that do not depend on temporal context.

What should attribution methods reveal about temporal processing? For models trained to predict neural responses to video, we want attribution methods to reveal: (1) which temporal windows in the input stimulus drive responses at specific output times, (2) how this temporal sensitivity adapts based on stimulus content (e.g., static vs. moving stimuli), and (3) whether learned temporal integration matches known biological phenomena like motion anticipation or predictive coding. Attribution methods can only reveal these properties if the model architecture can learn them in the first place.

6.2. *TL;DR*

I found failure to fit network and interpretative methods on time-series data in two ways:

First, my proposed Spatiotemporal Integrated Gradients (STIG) method mathematically reduces to standard integrated gradients due to the path independence theorem for gradient fields. Despite its intuitive appeal of sequentially revealing temporal information, STIG computes:

$$\mathrm{STIG}_{i,t} = (x_{i,t} - x'_{i,t}) \times \int_0^1 \frac{\partial f(x' + \beta(x - x'))}{\partial x_{i,t}} d\beta = \mathrm{IG}_{i,t} \tag{12}$$

This equivalence holds for any continuously differentiable function, regardless of the integration path. Consequently, no path-based attribution method can distinguish between models with true temporal dynamics and models with static temporal filters when applied to feedforward networks.

Second, Conv2D architectures with time encoded as input channels implement a fixed filter bank of spatiotemporal receptive fields. For these architectures, the gradient with respect to any temporal position is input-independent: $\frac{\partial y}{\partial x_t} = W_t$. Attribution thus reveals the architecture's learned biases—which temporal positions the fixed filters are

sensitive to—rather than how specific input dynamics drive specific output dynamics. Our empirical analysis (Figure X) demonstrates this limitation: different static images produce temporal response profiles with identical timing but different amplitudes for some units while having identical timing for other units, and neither captures the temporal structure of responses to dynamic stimuli.

6.3. *Architectural Requirements for Temporal Explainability*

These findings establish that meaningful temporal attribution requires architectures with three properties:

Temporal state maintenance. The model must carry information forward through time, enabling it to build representations that integrate over temporal context. This allows for phenomena like adaptation, prediction, and motion processing that require comparing current inputs to recent history.

6.4. *Implications for Computational Neuroscience and Embodied AI*

For our application domain—building biologically-constrained models of V1 that serve as vision encoders for quadruped robot control—these architectural requirements have direct implications. Biological visual cortex exhibits rich temporal dynamics including motion anticipation, omitted stimulus responses, and context-dependent adaptation. To model these phenomena and validate our models against neural data, we require architectures capable of learning such dynamics. Conv2D models with time-as-channels cannot learn these temporal computations, and consequently, attribution methods applied to these models cannot validate whether learned representations capture biological temporal processing.

Similarly, embodied robotic systems operating in dynamic environments require vision systems that process temporal information causally and adaptively. A robot navigating complex terrain must integrate visual motion information over time, predict future states based on temporal context, and adapt its visual processing based on behavioral goals—all requiring architectures with genuine temporal dynamics rather than static temporal filters.

6.5. *Future Directions*

This work establishes that the path forward for explainable temporal models lies in architectural innovation rather than attribution method development. Specific promising directions include:

Hybrid architectures combining 3D convolutions for hierarchical spatiotemporal feature extraction or networks with recurrent layers for temporal state maintenance may provide the right balance of biological plausibility and computational efficiency for modeling cortical dynamics. In the interest of simplicity and ease of implementation, we should switch our Conv2D models to Conv3D models and train our networks using the previously established pipeline.

Evaluating whether models with proper temporal architectures learn biologically-plausible temporal receptive fields compared to static filter bank models. This requires comparing learned temporal dynamics to known phenomena in retinal ganglion cells and V1 neurons responding to naturalistic stimuli.

Developing benchmark tasks that specifically test temporal causality, such as predicting neural responses to temporally manipulated stimuli (reversed sequences, frame-dropped sequences, or temporally jittered sequences) where models without true temporal processing should fail systematically.

Essentially, interpretability methods can reveal only what the model has actually learned. If we want to understand temporal processing in neural networks, we must first build architectures capable of learning temporal processing.

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B. AUTHORS' CONTRIBUTIONS

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C. CONFLICT OF INTEREST

The author declares no conflict of interest.

D. DATA AVAILABILITY

Empty

E. ETHICAL STATEMENT

Empty

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